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A Bayesian approach for non-homogeneous gamma degradation processes

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ABSTRACT

A Bayesian approach based on the Markov Chain Monte Carlo technique is proposed for the non-homogeneous gamma process with power-law shape function. Vague and informative priors, formalized on some quantities having a “physical” meaning, are provided. Point and interval estimation of process parameters and some functions thereof are developed, as well as prediction on some observable quantities that are useful in defining the maintenance strategy is proposed. Some useful approximations are derived for the conditional and unconditional mean and median of the residual life to reduce computational time. Finally, the proposed approach is applied to a real dataset.

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MATHEMATICS SUBJECT CLASSIFICATION

1. Introduction

The non-homogeneous gamma process is a widely accepted model to describe degradation phenomena when the degradation growth of a unit depends only on its age and not on the accumulated damage. Indeed, the non-homogeneous gamma process proved to be suitable to model a gradual damage that accumulates monotonically over time in a sequence of tiny increments, and it has been effectively adopted to describe wear, fatigue, corrosion, crack growth, erosion, and consumption. For further reading, refer to Abdel-Hameed (1975); Çinlar (1980); Singpurwalla (1985); Finkelstein (2003); Lawless and Crowder (2004), and, more recently, Mahmoodian and Alani (2014).

A continuous-time stochastic process $\{W(t); t \geq 0\}$ is said to be a gamma process with shape function $\eta(t)$ and scale parameter θ if it possesses the following properties:

1. $W(0) = 0$ with probability one;
2. the increment $\Delta W(t_1, t_2) = W(t_2) - W(t_1)$, for all $t_2 > t_1 \geq 0$, is distributed as a gamma probability density function (pdf) with shape parameter $\Delta\eta(t_1, t_2) = \eta(t_2) - \eta(t_1)$ and scale parameter θ :

$$f_{\Delta W}(\delta) = \frac{\delta^{\Delta\eta(t_1, t_2)-1}}{\theta^{\Delta\eta(t_1, t_2)} \Gamma[\Delta\eta(t_1, t_2)]} \exp(-\delta/\theta), \quad (1)$$

where $\eta(t)$ is a non-decreasing, right-continuous, real-valued function for $t \geq 0$, with $\eta(0) = 0$;

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3. $W(t)$ has independent increments over disjoint intervals.

If $\eta(t)$ is linear with t , say $\eta(t) = \lambda t$, the gamma process is said to be homogeneous since it is a stationary process, while it is a non-homogeneous process in the general case. In accordance with the definition of the gamma process, the mean and variance of the degradation $W(t)$ at time t are $E\{W(t)\} = \theta\eta(t)$ and $V\{W(t)\} = \theta^2\eta(t)$, respectively. It follows that the variance-to-mean ratio equals the scale parameter θ and therefore does not depend on time t .

Statistical estimation methods that were developed for the gamma process during the last decades include the maximum likelihood (ML) method and the method of moments, as well as the Bayesian method (see, e.g., van Noortwijk (2009) for a comprehensive survey). More recently, semiparametric estimation procedures, that avoids the need to formulate the functional form of $\eta(t)$, were proposed by Ye et al. (2014).

Although Bayesian methods enable the analysts to introduce in the inferential procedure technological information on the observed degradation phenomenon, as well as to make prediction on observable quantities in a simpler manner than the classical estimation procedures, little attention has been paid to Bayesian methods for degradation gamma processes in the literature.

In particular, Dufresne, Gerber, and Shiu (1991) proposed a Bayesian estimator of the parameter $u = 1/\theta$ of a homogeneous gamma process with linear shape function $\eta(t) = \lambda t$, under the assumption that the other parameter λ is known. Later, Kallen and van Noortwijk (2005) recast the homogeneous gamma process in terms of the coefficient of variation $\nu = 1/\sqrt{\lambda}$, that they assumed to be known, and the average deterioration rate $\mu = \lambda\theta$, that is left to be assessed. An inverted gamma density is then used to describe the prior information on the rate μ . The same assumption that only the rate μ is unknown was made successively by Lee, Kim, and Lee (2014), who selected, among a finite set of suitable distributions, the normal distribution to elicit the empirical prior information on the degradation rate μ . A Markov Chain Monte Carlo (MCMC) technique, based on the Metropolis-Hastings algorithm, was used to estimate the parameters. The assumption that only one parameter of the homogeneous gamma process is unknown was relaxed by Bousquet et al. (2015); they assumed that both the parameters, λ and θ , are unknown and a priori independent, and used both a non-informative prior density, say $g(\lambda, \theta) \propto \sqrt{\lambda}\Psi(\lambda) - 1/\theta$, where $\Psi(\lambda)$ is the digamma function and a joint informative prior given by the product of a gamma prior on λ and a (conditional) inverse gamma prior on θ , given λ . The MCMC technique was then used to make inference.

Finally, Bayesian estimators of the parameters of a non-homogeneous gamma processes were proposed by van Noortwijk (2009), under the assumption that the shape function is power-law, say $\eta(t) = ct^\beta$. The power-law parameter β is there assumed to be known and the joint prior distributions on the parameters $u = 1/\theta$ and c are assigned under the assumption that u , given c , is a priori gamma distributed.

Thus, this paper is focused on the non-homogeneous gamma process with power-law $\eta(t)$ function, rewritten as $\eta(t) = (t/\alpha)^\beta$, that was shown to describe adequately several degradation phenomena, such as degradation due to sulphate attack, diffusion-controlled ageing, and the expected scour-hole depth (see, e.g., van Noortwijk 2009).

At first, the paper provides some novel distributional approximations of the conditional and unconditional remaining life that can be usefully exploited both in a Bayesian context, and when classical estimation procedures, such as the maximum likelihood one, are adopted.

Then, in order to fill in some gaps in existing literature, this paper proposes a Bayesian procedure, based on the MCMC technique, for the non-homogeneous gamma process with power-law $\eta(t)$ function, under the assumption that all parameters are unknown.

Therefore, it is assumed that prior information is available on quantities having a physical meaning and hence an immediate interpretation for the analyst. In particular, it is assumed that the analyst is able to formulate prior information on: *a*) the shape (convex or concave) of the mean degradation curve, *b*) the mean degradation level at a given time, and *c*) the (time-constant) ratio between the variance and the mean of the degradation process. Both vague and informative priors are proposed and discussed, that allow different levels of technological knowledge to be introduced in the inferential procedure. Posterior inference is made on the process parameters and on several functions thereof, such as the mean degradation level at an arbitrary time and the residual reliability, by using MCMC techniques. Prediction of the degradation increment over a future time interval and of the remaining life is also provided. Finally, the proposed procedure is applied to a real dataset given in Meeker and Escobar (1998) and more recently analyzed by Ebrahem and Higgins (2006), that consists of the sliding wear data of four metal alloy specimens.

2. Distributional approximations of the remaining life

A quantity of primary interest for the lifetime analysis of a technological unit is the remaining life X , typically used to plan the maintenance of a degrading unit that is unfailed at the current time τ . For units subject to degradation phenomena, the failure is assumed to occur when the degradation level exceed the threshold level $w_{\max}$. Then, the remaining life X is the further operating time the unit spends until failure, when starting from the current state $w_\tau = W(\tau) \leq w_{\max}$. The exact cumulative distribution function (Cdf) of X , given the current degradation level w_τ , is:

$$\begin{aligned} F_{X|w_\tau}(x) &= \Pr\{X \leq x | W(\tau) = w_\tau\} = \Pr\{\Delta W(\tau, \tau + x) > w_{\max} - w_\tau\} \\ &= 1 - \text{IG}\left[\frac{w_{\max} - w_\tau}{\theta}; \Delta\eta(\tau, \tau + x)\right], \end{aligned} \quad (2)$$

where $\text{IG}[z; a]$ denote the gamma Cdf with unit scale parameter and shape parameter a . The median of the conditional remaining life, say $X_{0.5|w_\tau}$, is the (numerical) solution of $\text{IG}[(w_{\max} - w_\tau)/\theta; \Delta\eta(\tau, \tau + X_{0.5|w_\tau})] = 0.5$, and the pdf of X , say $f_{X|w_\tau}(x)$, is given by the derivative of (2) with respect to x , which is not in closed-form and must be computed numerically.

A simple closed-form approximation for $f_{X|w_\tau}(x)$ can be derived when $\eta(t) = (t/\alpha)^\beta$, with $\alpha, \beta > 0$, by extending the results obtained by Paroissin and Salami (2014) for the life-time distribution. In particular, we assume that the random variable $Z = \Delta\eta(\tau, \tau + X) = [(\tau + X)/\alpha]^\beta - (\tau/\alpha)^\beta$, given that $W(\tau) = w_\tau$, is approximately distributed as the Birnbaum-Saunders (BS) distribution (Birnbaum and Saunders 1969) with parameters $\alpha^* = \sqrt{\theta/(w_{\max} - w_\tau)}$ and $\beta^* = (w_{\max} - w_\tau)/\theta \equiv (1/\alpha^*)^2$:

$$F_{Z|w_\tau}(z) \cong \Phi\left[\sqrt{\frac{w_{\max} - w_\tau}{\theta}}\left(\sqrt{\frac{z}{(w_{\max} - w_\tau)/\theta}} - \sqrt{\frac{(w_{\max} - w_\tau)/\theta}{z}}\right)\right], \quad (3)$$

where $\Phi(y)$ denotes the standard Normal Cdf. Note that the approximation (3) implies that the degradation increments over disjoint time intervals are independent random variables with finite mean and variance, as it actually occurs under the non-homogeneous gamma process. Thus, from (3), by making the change of variables $Z = [(\tau + X)/\alpha]^\beta - (\tau/\alpha)^\beta$, the pdf

of the conditional remaining life X is, approximately, given by:

$$f_{X|w_\tau}(x) \cong \frac{1}{2\sqrt{2\pi}} \sqrt{\frac{w_{\max} - w_\tau}{\theta}} \frac{\beta(\tau + x)^{\beta-1}}{(\tau + x)^\beta - \tau^\beta} \left(\sqrt{\frac{[(\tau + x)/\alpha]^\beta - (\tau/\alpha)^\beta}{(w_{\max} - w_\tau)/\theta}} + \sqrt{\frac{(w_{\max} - w_\tau)/\theta}{[(\tau + x)/\alpha]^\beta - (\tau/\alpha)^\beta}} \right) \times \exp \left[-\frac{w_{\max} - w_\tau}{2\theta} \left(\sqrt{\frac{[(\tau + x)/\alpha]^\beta - (\tau/\alpha)^\beta}{(w_{\max} - w_\tau)/\theta}} - \sqrt{\frac{(w_{\max} - w_\tau)/\theta}{[(\tau + x)/\alpha]^\beta - (\tau/\alpha)^\beta}} \right)^2 \right]. \quad (4)$$

It is easy to note that the pdf (4) depends on both the current age τ of the unit and its current degradation level w_τ .

From (2), the exact mean of the conditional remaining life is given by

$$E\{X|w_\tau\} = \int_0^\infty \text{IG}[(w_{\max} - w_\tau)/\theta; \Delta\eta(\tau, \tau + x)] dx, \quad (5)$$

and its computation requires a numerical integration. Then, an approximate simpler expression can be attractive. To obtain it, we note that the mean and variance of the BS variable $Z|w_\tau$ are equal to $\beta^*[\alpha^{*2} + 2]/2$ and $(\beta^*\alpha^*)^2(5\alpha^{*2} + 4)/4$, respectively, (Birnbaum and Saunders 1969) and hence, from (3), the (conditional) mean and variance of $Z|w_\tau$ are approximately given by $E\{Z|w_\tau\} \cong (w_{\max} - w_\tau)/\theta + 1/2$ and $\text{Var}\{Z|w_\tau\} \cong (w_{\max} - w_\tau)/\theta + 5/4$, respectively. Then, because $x = x(z) = \alpha[z + (\tau/\alpha)^\beta]^{1/\beta} - \tau$ and

$$\begin{aligned} \frac{d^2x}{dz^2} \Big|_{E\{Z|w_\tau\}} &= \frac{\alpha(1-\beta)}{\beta^2} \left[E\{Z|w_\tau\} + \left(\frac{\tau}{\alpha}\right)^\beta \right]^{1/\beta-2} \\ &\cong \frac{\alpha(1-\beta)}{\beta^2} \left[\frac{w_{\max} - w_\tau}{\theta} + \frac{1}{2} + \left(\frac{\tau}{\alpha}\right)^\beta \right]^{1/\beta-2}, \end{aligned}$$

by using the delta method (see, e.g., Eq. (3) in Oehlert (1992) and Eq. (4) in Ver Hoef (2012)), the “BS approximate” mean of the conditional remaining life is given by:

$$\begin{aligned} E\{X|w_\tau\} &\cong x(z)|_{E\{Z|w_\tau\}} + \frac{1}{2} \text{Var}\{Z|w_\tau\} \cdot \frac{d^2x}{dz^2} \Big|_{E\{Z|w_\tau\}} \\ &= \alpha \left[\frac{w_{\max} - w_\tau}{\theta} + \frac{1}{2} + \left(\frac{\tau}{\alpha}\right)^\beta \right]^{1/\beta} - \tau + \frac{\alpha(1-\beta) \left(\frac{w_{\max} - w_\tau}{\theta} + \frac{5}{4} \right)}{2\beta^2 \left[\frac{w_{\max} - w_\tau}{\theta} + \frac{1}{2} + \left(\frac{\tau}{\alpha}\right)^\beta \right]^{2-1/\beta}} \quad (6) \end{aligned}$$

Since the scale parameter β^* represents the median of the (approximate) BS distribution (3), then the median of the conditional remaining life $X|w_\tau$ is approximately given by:

$$X_{0.5}|w_\tau \cong \alpha \left[(w_{\max} - w_\tau)/\theta + (\tau/\alpha)^\beta \right]^{1/\beta} - \tau. \quad (7)$$

When one has to plan the maintenance of a population of units survived up to a given time τ , it is useful to predict the unconditional remaining life, which is not affected by the degradation level $w_\tau = W(\tau)$, $w_\tau \leq w_{\max}$, reached by the unit at τ . Its Cdf is given by

$$\begin{aligned}
 F_X(x) &= \Pr\{\tau < T \leq \tau + x | T > \tau\} = [F_T(\tau + x) - F_T(\tau)] / [1 - F_T(\tau)] \\
 &= 1 - \frac{\text{IG}\left[w_{\max}/\theta; ((\tau + x)/\alpha)^\beta\right]}{\text{IG}\left[w_{\max}/\theta; (\tau/\alpha)^\beta\right]}, \quad x \geq 0,
 \end{aligned} \quad (8)$$

where $F_T(t) = \Pr\{W(t) > w_{\max}\}$ is the Cdf of the lifetime T .

The derivative of (8) has the same computational difficulties as the derivative of the conditional Cdf (2) and hence an approximation for $F_X(x)$, that allows the unconditional pdf $f_X(x)$ to be expressed in a tractable (although approximate) form, is desirable. At this aim, we first obtain from (3) an approximate expression of $F_T(t)$ by making the change of variables $Z = (T/\alpha)^\beta$ and setting $w_\tau = 0$:

$$F_T(t) = 1 - \text{IG}\left[\frac{w_{\max}}{\theta}; \left(\frac{t}{\alpha}\right)^\beta\right] \cong \Phi\left[\sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{(t/\alpha)^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{(t/\alpha)^\beta}}\right)\right]. \quad (9)$$

Hence, from (9), the desired approximate Cdf of the unconditional remaining life $X = T - \tau$ of the units that are unfailed at τ , irrespective of the degradation level w_τ accumulated at τ , results in

$$\begin{aligned}
 F_X(x) &\cong \frac{\Phi\left[\sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{[(\tau+x)/\alpha]^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{[(\tau+x)/\alpha]^\beta}}\right)\right] - \Phi\left[\sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{(\tau/\alpha)^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{(\tau/\alpha)^\beta}}\right)\right]}{1 - \Phi\left[\sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{(\tau/\alpha)^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{(\tau/\alpha)^\beta}}\right)\right]}
 \end{aligned} \quad (10)$$

Therefore, the approximate unconditional pdf can be derived by differencing (10):

$$\begin{aligned}
 f_X(x) &\cong \frac{1}{1 - F_T(\tau)} \cdot \frac{\beta}{2(\tau + x)\sqrt{2\pi}} \sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{[(\tau + x)/\alpha]^\beta}{w_{\max}/\theta}} + \sqrt{\frac{w_{\max}/\theta}{[(\tau + x)/\alpha]^\beta}} \right) \\
 &\times \exp\left[-\frac{1}{2} \frac{w_{\max}}{\theta} \left(\sqrt{\frac{[(\tau + x)/\alpha]^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{[(\tau + x)/\alpha]^\beta}} \right)^2\right], \quad x \geq 0,
 \end{aligned} \quad (11)$$

where

$$1 - F_T(\tau) \cong 1 - \Phi\left[\sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{(\tau/\alpha)^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{(\tau/\alpha)^\beta}}\right)\right]$$

is the probability that the unit is actually unfailed at τ .

“BS approximate” expressions for the mean and median of the unconditional remaining life, say $E\{X\}$ and $X_{0.5}$, respectively, can be easily obtained by observing from (10) that the random variable:

$$Y = \sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{[(\tau + X)/\alpha]^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{[(\tau + X)/\alpha]^\beta}} \right), \quad Y > \kappa \quad (12)$$

is approximately a one-sided truncated standard Normal variable, whose truncation point is

$$\kappa = \sqrt{\frac{w_{\max}}{\theta}} \left(\sqrt{\frac{(\tau/\alpha)^\beta}{w_{\max}/\theta}} - \sqrt{\frac{w_{\max}/\theta}{(\tau/\alpha)^\beta}} \right).$$

Its mean and variance are then approximately given by:

$$E\{Y|Y > \kappa\} = \frac{\phi(\kappa)}{1 - \Phi(\kappa)} \quad \text{and} \quad \text{Var}\{Y|Y > \kappa\} = 1 - \frac{\phi(\kappa)}{1 - \Phi(\kappa)} \left[\frac{\phi(\kappa)}{1 - \Phi(\kappa)} - \kappa \right]$$

respectively, where $\phi(y)$ denotes the standard Normal pdf (Greene 2003). From (12), we have that

$$x(y) = \alpha \left\{ \frac{w_{\max}}{\theta} \left(y \sqrt{\frac{\theta}{4w_{\max}}} + \sqrt{\frac{y^2\theta}{4w_{\max}} + 1} \right)^2 \right\}^{1/\beta} - \tau. \quad (13)$$

Then, by using the delta method, the “BS approximate” mean of the unconditional remaining life is:

$$E\{X\} \cong x(y)|_{E\{Y|Y>\kappa\}} + \frac{1}{2} \text{Var}\{Y|Y > \kappa\} \cdot \frac{d^2x}{dy^2} \Big|_{E\{Y|Y>\kappa\}} \quad (14)$$

where

$$\frac{d^2x}{dy^2} = \frac{2\alpha\theta}{\beta^2 w_{\max}} \left(2\sqrt{\frac{y^2\theta}{w_{\max}} + 4} - \beta y \sqrt{\frac{\theta}{w_{\max}}} \right) \left(\sqrt{\frac{y^2}{4} + \frac{w_{\max}}{\theta}} + \frac{y}{2} \right)^{2/\beta} \left(\frac{y^2\theta}{w_{\max}} + 4 \right)^{-3/2}.$$

Since the median of the truncated standard Normal distribution with truncation point κ is equal to $Y_{0.5} = \Phi^{-1}[1/2 + \Phi(\kappa)/2]$, where $\Phi^{-1}(s)$ is the s -th percentile of the standard Normal distribution, the approximate median of the unconditional remaining life $X_{0.5}$ can be easily obtained from (13) by using $Y_{0.5}$ in place of y :

$$X_{0.5} \cong \alpha \left\{ \frac{w_{\max}}{\theta} \left(\sqrt{\left\{ \Phi^{-1} \left[\frac{1 + \Phi(\kappa)}{2} \right] \right\}^2 \frac{\theta}{4w_{\max}}} + \sqrt{\left\{ \Phi^{-1} \left[\frac{1 + \Phi(\kappa)}{2} \right] \right\}^2 \frac{\theta}{4w_{\max}} + 1} \right)^2 \right\}^{1/\beta} - \tau \quad (15)$$

3. The Bayesian inferential procedure

Let us suppose that m systems operate under identical conditions, and that each system is inspected n_i times at possibly not equal ages $t_{i,k}$ ($k = 1, \dots, n_i$). Let $\delta_{i,k} = W(t_{i,k}) - W(t_{i,k-1})$ denote the degradation increment of system i during the k -th inspection interval $(t_{i,k-1}, t_{i,k})$, where $t_{i,0} = 0$ for all i . Then, the likelihood function relative to the observed data $\mathbf{w} = (\delta_{1,1}, \dots, \delta_{1,n_1}, \dots, \delta_{m,1}, \dots, \delta_{m,n_m})$ is given by:

$$L(\mathbf{w}) = \prod_{i=1}^m \prod_{k=1}^{n_i} \frac{\delta_{i,k}^{\eta(t_{i,k}) - \eta(t_{i,k-1}) - 1}}{\theta^{\eta(t_{i,k}) - \eta(t_{i,k-1})} \Gamma[\eta(t_{i,k}) - \eta(t_{i,k-1})]} \exp(-\delta_{i,k}/\theta). \quad (16)$$

Let us assume that $\eta(t) = (t/\alpha)^\beta$. We also assume that the analyst possesses some prior information on the degradation phenomenon under study, which arise from the observation or the knowledge of similar degradation phenomena. In particular, we assume that prior information is available on the following quantities having a physical meaning:

- the variance-to-mean ratio $\theta = \text{Var}\{W(t)\}/E\{W(t)\}$, which is constant under the gamma process;

- the behavior of the mean degradation curve $E\{W(t)\} = \theta(t/\alpha)^\beta$, which is: *a*) linear with time t when the power-law parameter β is equal to 1 (so that the process is homogeneous), *b*) downward concave when $\beta < 1$, and *c*) convex when $\beta > 1$. Thus, a prior information on the behavior of $E\{W(t)\}$ can be easily converted into a prior information on the shape parameter β , taking into account that the more the mean curve deviates from the linear trend, the more β is far from 1;
- the mean degradation level $E_\omega \equiv E\{W(\omega)\} = \theta(\omega/\alpha)^\beta$ at an arbitrary time ω .

Flexible prior pdf's on the variance-to-mean ratio θ and the mean degradation level E_ω , that are assumed to be prior independent of each other, are respectively:

- The inverse gamma distribution $\text{IGamma}(a, b)$ on the ratio θ :

$$g(\theta) = \frac{b^a \theta^{-a-1}}{\Gamma(a)} \exp(-b/\theta), \quad a, b > 0, \quad (17)$$

whose parameters a and b can be easily obtained once prior values to the mean and variance of θ , say $E\{\theta\} = b/(a-1)$ and $V\{\theta\} = b^2/[(a-1)^2(a-2)]$, are formulated, so that $a = E^2\{\theta\}/V\{\theta\} + 2$ and $b = E\{\theta\}[E^2\{\theta\}/V\{\theta\} + 1]$. If no information on θ is available, the (improper) vague prior $g(\theta) = 1/\theta$ is used, which can be viewed as a special case of (17) for $a = b = 0$.

- The gamma distribution $\text{Gamma}(c, d)$ on the mean degradation level E_ω :

$$g(E_\omega) = \frac{E_\omega^{c-1}}{d^c \Gamma(c)} \exp(-E_\omega/d), \quad c, d > 0 \quad (18)$$

whose parameters c and d are easily obtained once prior values to the mean and variance of E_ω , say $E\{E_\omega\} = cd$ and $V\{E_\omega\} = cd^2$, are formulated, so that $c = E^2\{E_\omega\}/V\{E_\omega\}$ and $d = V\{E_\omega\}/E\{E_\omega\}$. If no information on E_ω is available, then the (improper) vague prior $g(E_\omega) = 1/E_\omega$ is to be used, which can be viewed as a special case of (18) for $c = 0$ and $d \rightarrow \infty$.

Referring to β , we suggest the following prior distributions depending on the prior information available on the behavior of the mean degradation curve $E\{W(t)\}$:

- The (improper) vague prior $g(\beta) \propto 1/\beta$ when no information on $E\{W(t)\}$ is available;
- The 1-parameter gamma distribution $\text{Gamma}(p)$:

$$g(\beta) = \frac{p^p \beta^{p-1}}{\Gamma(p)} \exp(-\beta p) \quad (19)$$

with mean equal to 1, when the mean curve is known to be linear or almost linear. The parameter p is obtained as $p = 1/V\{\beta\}$, once a prior variance $V\{\beta\}$ on β is formulated;

- The Beta distribution $\text{Beta}(p, q)$:

$$g(\beta) = \frac{\beta^{p-1}(1-\beta)^{q-1}}{B(p, q)}, \quad 0 \leq \beta \leq 1, \quad p, q > 0 \quad (20)$$

where $B(p, q)$ denote the Beta function, when the mean curve is known to be concave. The Beta parameters p and q are obtained once a prior mean and variance on β , say $E\{\beta\} = p/(p+q)$ and $V\{\beta\} = pq/[(p+q)^2(p+q+1)]$, are formulated, so that $p = E^2\{\beta\}/V\{\beta\}(1-E\{\beta\}) - E\{\beta\}$ and $q = p/E\{\beta\} - p$. If the only available information is that $\beta < 1$, then the parameters of the Beta prior (20) can be both set equal to 0.5;

- The 3-parameter gamma distribution $\text{Gamma}(p, q, 1)$ with location parameter one:

$$g(\beta) = \frac{(\beta-1)^{p-1}}{q^p \Gamma(p)} \exp[-(\beta-1)/q], \quad \beta > 1, \quad p, q > 0 \quad (21)$$

when the mean curve is known to be convex. The parameters p and q are obtained once prior values to the mean and variance of β , say $E\{\beta\} = pq + 1$ and $V\{\beta\} = pq^2$, are formulated, so that $q = V\{\beta\}/(E\{\beta\} - 1)$ and $p = (E\{\beta\} - 1)/q$. If the only available information is that $\beta > 1$, then the improper vague prior $g(\beta) \propto 1/(\beta - 1)$ is to suggested, which can be viewed as a special case of (21) for $p = 0$ and $q \rightarrow \infty$.

Parameter β , too, is assumed to be prior independent of θ and E_ω , and hence the joint prior pdf on the quantities $(\beta, \theta, E_\omega)$ is given by $g(\beta, \theta, E_\omega) = g(\beta)g(\theta)g(E_\omega)$.

We use here the term “vague” for priors to denote diffuse prior distributions which do not require the formulation of a substantial prior knowledge on the specific quantity.

The likelihood function (16) can be easily recast in terms of the prior quantities $(E_\omega, \beta, \theta)$, by making the change of parameters $\alpha = \omega(\theta/E_\omega)^{1/\beta}$:

$$\begin{aligned} L(\mathbf{w}; E_\omega, \beta, \theta) &= \prod_{i=1}^m \prod_{j=1}^{n_i} \frac{\delta_{i,j}^{(E_\omega/\theta)[(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] - 1} \exp(-\delta_{i,k}/\theta)}{\theta^{(E_\omega/\theta) \sum_{i=1}^m (T_i/\omega)^\beta} \Gamma \left\{ \left(\frac{E_\omega}{\theta} \right) \left[\left(\frac{t_{i,k}}{\omega} \right)^\beta - \left(\frac{t_{i,k-1}}{\omega} \right)^\beta \right] \right\}} \\ &= \frac{1}{\theta^{(E_\omega/\theta) \sum_{i=1}^m (T_i/\omega)^\beta}} \prod_{i=1}^m \prod_{k=1}^{n_i} \frac{\delta_{i,k}^{(E_\omega/\theta)[(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] - 1} \exp[-w_{TOT}/\theta]}{\Gamma \left\{ (E_\omega/\theta) [(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] \right\}} \end{aligned} \quad (22)$$

where $T_i \equiv t_{i,n_i}$ and $w_{TOT} \equiv \sum_{i=1}^m \sum_{k=1}^{n_i} \delta_{i,k}$ are the last inspection time of unit i and the total measured wear, respectively. Given the joint prior pdf $g(\theta, \beta, E_\omega)$, the resulting joint posterior pdf is:

$$\pi(E_\omega, \beta, \theta | \mathbf{w}) = \frac{1}{D} \cdot \frac{g(\theta, \beta, E_\omega)}{\theta^{(E_\omega/\theta) \sum_{i=1}^m (T_i/\omega)^\beta}} \times \prod_{i=1}^m \prod_{k=1}^{n_i} \frac{\delta_{i,k}^{(E_\omega/\theta)[(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] - 1} \exp[-w_{TOT}/\theta]}{\Gamma \left\{ (E_\omega/\theta) [(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] \right\}} \quad (23)$$

where the denominator D is given by:

$$\begin{aligned} D &= \frac{b^a}{\Gamma(a)} \frac{1}{d^c \Gamma(c)} \int_\theta \int_\beta \int_{E_\omega} g(\beta) \frac{E_\omega^{c-1}}{\theta^{a+1}} \exp\left(-\frac{E_\omega}{d}\right) \frac{\exp[-(w_{TOT} + b)/\theta]}{\theta^{(E_\omega/\theta) \sum_{i=1}^m (T_i/\omega)^\beta}} \\ &\quad \times \prod_{i=1}^m \prod_{k=1}^{n_i} \frac{\delta_{i,k}^{(E_\omega/\theta)[(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] - 1}}{\Gamma \left\{ (E_\omega/\theta) [(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] \right\}} dE_\omega d\beta d\theta \end{aligned}$$

From (23), by making the change of variable $E_\omega = \theta(\omega/\alpha)^\beta$, the joint posterior pdf on the “original” gamma parameters (α, β, θ) can be obtained:

$$\begin{aligned} \pi(\alpha, \beta, \theta | \mathbf{w}) &= \frac{1}{D} \frac{b^a}{\Gamma(a)} \frac{1}{d^c \Gamma(c)} g(\beta) \frac{\beta}{\alpha} \left(\frac{\omega}{\alpha} \right)^{\beta c} \exp \left[-\frac{\theta}{d} \left(\frac{\omega}{\alpha} \right)^\beta \right] \\ &\quad \times \frac{\exp[-(w_{TOT} + b)/\theta]}{\theta \sum_{i=1}^m (T_i/\alpha)^\beta + a - c + 1} \prod_{i=1}^m \prod_{k=1}^{n_i} \frac{\delta_{i,j}^{[(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] - 1}}{\Gamma \left\{ (t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta \right\}}, \end{aligned} \quad (24)$$

from which the posterior pdf of any process parameter can be derived. For example, the marginal posterior pdf on α results in:

$$\pi(\alpha | \mathbf{w}) = \int_\theta \int_\beta \pi(\alpha, \beta, \theta | \mathbf{w}) d\beta d\theta$$

and the posterior mean is:

$$E\{\alpha|\mathbf{w}\} = \int_0^\infty \alpha \cdot \pi(\alpha|\mathbf{w}) d\alpha$$

Both the denominator of the posterior pdf and the numerator of the posterior mean of each parameter require the numerical computation of a tri-variate integral, whereas the numerator of the marginal posterior pdf's requires a bivariate integration. For instance, by making in (11) the change of variable $E_t = (t/\alpha)^\beta \theta$, the marginal posterior pdf on the mean degradation level E_t at any arbitrary time t can be obtained:

$$\begin{aligned} \pi(E_t|\mathbf{w}) &= \frac{1}{D} \frac{b^a}{\Gamma(a)} \frac{1}{d^c \Gamma(c)} \int_\theta \int_\beta g(\beta) E_t^{c-1} \left(\frac{\omega}{t}\right)^{\beta c} \exp\left[-\frac{E_t}{d} \left(\frac{\omega}{t}\right)^\beta\right] \\ &\times \frac{\exp[-(w_{TOT} + b)/\theta]}{\theta^{(E_t/\theta) \sum_{i=1}^m (T_i/t)^\beta + a+1}} \prod_{i=1}^m \prod_{k=1}^{n_i} \frac{\delta_{i,k}^{(E_t/\theta) [(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta] - 1}}{\Gamma\{(E_t/\theta) [(t_{i,k}/\omega)^\beta - (t_{i,k-1}/\omega)^\beta]\}} d\beta d\theta, \end{aligned} \quad (25)$$

and its posterior mean is

$$E\{E_t|\mathbf{w}\} = \int_0^\infty E_t \cdot \pi(E_t|\mathbf{w}) dE_t.$$

The same procedure can be used for obtaining the marginal posterior pdf and mean of any other function of the process parameters, for instance: (a) the conditional residual reliability

$$R_\tau(x|w_\tau) = 1 - F_{X|w_\tau}(x) = \text{IG}\left[\frac{w_{\max} - w_\tau}{\theta}; \Delta\eta(\tau, \tau + x)\right], \quad (26)$$

given the degradation level $w_\tau = W(\tau)$ at the current time τ , where $\Delta\eta(\tau, \tau + x) = [(\tau + x)/\alpha]^\beta - (\tau/\alpha)^\beta$, and (b) the unconditional residual reliability

$$R_\tau(x) = 1 - F_X(x) = \frac{\text{IG}\left[\frac{w_{\max}}{\theta}; \left(\frac{\tau+x}{\alpha}\right)^\beta\right]}{\text{IG}\left[\frac{w_{\max}}{\theta}; \left(\frac{\tau}{\alpha}\right)^\beta\right]}. \quad (27)$$

The Bayesian procedure allows also to easily predict any observable quantity. For example, the posterior predictive pdf of the degradation increment $\Delta W = W(T_i + \Delta t) - W(T_i)$ of unit i during the future time interval $(T_i, T_i + \Delta t)$ is given by:

$$\begin{aligned} f_{\Delta W}(\delta|\mathbf{w}) &= \int_\theta \int_\beta \int_\alpha \pi(\alpha, \beta, \theta|\mathbf{w}) f_{\Delta W}(\delta) d\alpha d\beta d\theta \\ &= \int_\theta \int_\beta \int_\alpha \pi(\alpha, \beta, \theta|\mathbf{w}) \frac{\delta^{\Delta\eta(T_i, T_i + \Delta t) - 1} \exp(-\delta/\theta)}{\theta^{\Delta\eta(T_i, T_i + \Delta t)} \Gamma[\Delta\eta(T_i, T_i + \Delta t)]} d\alpha d\beta d\theta, \end{aligned} \quad (28)$$

where T_i is the last inspection time of unit i and $\Delta\eta(T_i, T_i + \Delta t) = [(T_i + \Delta t)/\alpha]^\beta - (T_i/\alpha)^\beta$. The posterior mean of the degradation increment is:

$$E\{\Delta W|\mathbf{w}\} = \int_\theta \int_\beta \int_\alpha \pi(\alpha, \beta, \theta|\mathbf{w}) \theta \left[\left(\frac{T_i + \Delta t}{\alpha}\right)^\beta - \left(\frac{T_i}{\alpha}\right)^\beta \right] d\alpha d\beta d\theta, \quad (29)$$

where $\theta\{[(T_i + \Delta t)/\alpha]^\beta - (T_i/\alpha)^\beta\}$ is the mean increment $E\{\Delta W\}$. In a similar manner, by using in (28) the predictive pdf (4) or (11) in place of the predictive pdf $f_{\Delta W}(\delta)$, the approximate posterior predictive pdf of the conditional or unconditional remaining

life can be obtained, respectively. Likewise, the conditional and unconditional posterior mean of X can be approximated by using (6) and (14), respectively, in place of $E\{\Delta W\} = \theta\{[(T_i + \Delta t)/\alpha]^\beta - (T_i/\alpha)^\beta\}$ in (29).

4. The Markov Chain Monte Carlo procedure

The Bayesian inferential procedure presented in Section 3 could be implemented, in line of principle, by adopting numerical multivariate integration, for instance a tri-variate integration is necessary to compute the denominator D in (23), or the posterior predictive pdf of the degradation increment (28), or the expected value in (29). A quadri-variate integral is instead required to estimate the mean remaining life, if you do not want to use the approximations (6) and (14). However, numerical multidimensional integration is often unfeasible or highly time consuming in the practice. Thus, in this paper we adopt a MCMC technique in order to reduce the computational burden and thus the execution time of the computer code.

At this aim, we use OpenBUGS (Lunn et al. 2009), and the slice sampling algorithm (Neal 2003) is here adopted because it typically provides better performance with respect to the Metropolis-Hastings that requires a selection of an adequate proposal distribution and a careful tuning of the step-size parameters. We draw a three-dimensional vector sample of size N , say $(E_{\omega_j}, \beta_j, \theta_j)$, $j = 1, \dots, N$, from the posterior pdf $\pi(E_\omega, \beta, \theta|\mathbf{w})$ in (23), generated after a sufficiently large burn-in period performed to make negligible the influence of the starting point of the numerical procedure. Convergence to the stationary (target) distribution $\pi(E_\omega, \beta, \theta|\mathbf{w})$ of the Markov Chain is also monitored.

The posterior vector sample $(\alpha_j, \beta_j, \theta_j)$, $j = 1, \dots, N$, of the “original” gamma parameters can be easily derived by noting that $\alpha = \omega(\theta/E_\omega)^{1/\beta}$. Thus, the posterior mean and $(1 - \gamma)$ highest posterior density (HPD) interval of each parameter can be estimated: the former is given by the mean of the corresponding elements of the posterior sample, for instance $E\{\alpha|\mathbf{w}\} = \int_0^\infty \alpha \cdot \pi(\alpha|\mathbf{w})d\alpha \cong \sum_{j=1}^N \alpha_j/N$, while the latter is obtained by ordering the posterior sample and selecting the shortest interval containing the fraction $(1 - \gamma)$ of the sample.

From the posterior vector sample $(\alpha_j, \beta_j, \theta_j)$, $j = 1, \dots, N$, the posterior sample of any function of the gamma parameters, say $h = h(\alpha, \beta, \theta)$, is given by $h_i = h(\alpha_i, \beta_i, \theta_i)$, $i = 1, \dots, N$, from which the posterior pdf, the mean, the median and the HPD interval of such a quantity are easily obtained (Robert and Casella 2010).

By using the gamma distribution of the degradation increment Δw and the Cdf's (2) and (8) of the conditional and unconditional remaining life, the posterior samples of Δw , $X|w_\tau$, and X are generated. From those predictive posterior samples, the posterior means and medians are easily obtained. However, inverting the exact Cdf's (2) and (8) for drawing the pseudo-random samples of $X|w_\tau$ and X is computationally very onerous.

On the contrary, a considerable reduction of the computation time is obtained by using the approximate Cdf (3) to generate approximate posterior pseudo-random samples of the conditional remaining life $X|w_\tau$. In particular, given the current degradation level w_τ , the conditional remaining life sample $x_j|w_\tau$, $j = 1, \dots, N$, is approximated by firstly generating a sample of size N , say $z_j|w_\tau$, $j = 1, \dots, N$, from the BS Cdf (3) with parameters $\alpha_j^* = \sqrt{\theta_j/(w_{\max} - w_\tau)}$ and $\beta_j^* = (1/\alpha_j^*)^2$, and then transforming each element $z_j|w_\tau$ by

$$x_j|w_\tau = \alpha_j[z_j|w_\tau + (\tau/\alpha_j)^{\beta_j}]^{1/\beta_j} - \tau, \quad j = 1, \dots, N \quad (30)$$

From the posterior sample (30), the posterior predictive pdf $f_{X|w_\tau}(x|\mathbf{w})$, the posterior mean and median (that will be called the “BS exact” mean and median), and the $(1 - \gamma)$

HPD interval of the conditional remaining life are easily derived. Alternatively, the posterior mean $E\{X|w_\tau, \mathbf{w}\}$ can be estimated by using the approximation (6) and treating the posterior mean as a function of the model parameters, so that the “BS approximate” posterior mean results in $E\{X|w_\tau, \mathbf{w}\} \cong \sum_{j=1}^N E\{X|w_\tau, \alpha_j, \beta_j, \theta_j\}/N$, where $(\alpha_j, \beta_j, \theta_j)$ is the posterior vector sample. This approximation makes the generation of the BS sample of the $z_j|w_\tau$ and the successive transformation (30) unnecessary, thus reducing in principle the computation time.

A little more complex and time onerous is the generation of the approximate posterior pseudo-random sample x_j , $j = 1, \dots, N$, of the unconditional remaining life X , because a sample can not be directly generated by OpenBUGS from the Cdf (10). In particular, for each j , given the current time τ , we first estimate the Normal Cdf

$$\Phi_j = \Phi \left[\sqrt{\frac{w_{\max}}{\theta_j}} \left(\sqrt{\frac{(\tau/\alpha_j)^{\beta_j}}{w_{\max}/\theta_j}} - \sqrt{\frac{w_{\max}/\theta_j}{(\tau/\alpha_j)^{\beta_j}}} \right) \right]$$

and draw a pseudo-random element u_j from the Uniform distribution $U(0, 1)$. Then, x_j is the (positive) solution of

$$\Phi \left[\sqrt{\frac{w_{\max}}{\theta_j}} \left(\sqrt{\frac{[(\tau + x_j)/\alpha_j]^{\beta_j}}{w_{\max}/\theta_j}} - \sqrt{\frac{w_{\max}/\theta_j}{[(\tau + x_j)/\alpha_j]^{\beta_j}}} \right) \right] = u_j (1 - \Phi_j) + \Phi_j = C_j,$$

that is

$$x_j = \alpha_j y_j^{2/\beta_j} - \tau, \quad j = 1, \dots, N \quad (31)$$

where

$$y_j = \left(\Phi^{-1}(C_j) + \sqrt{[\Phi^{-1}(C_j)]^2 + 4w_{\max}/\theta_j} \right) / 2$$

and $\Phi^{-1}(\cdot)$ denotes the inverse standard Normal Cdf. From the posterior sample x_j , $j = 1, \dots, N$, the posterior predictive pdf $f_X(x|\mathbf{w})$, the “BS exact” posterior mean and median, and the $(1 - \gamma)$ HPD interval of the unconditional remaining life are easily derived.

If only the unconditional posterior mean $E\{X|\mathbf{w}\}$ is to be estimated, and not the posterior distribution of X , the execution time can be drastically reduced by using the approximation (14) and treating the posterior mean as a function of the model parameters, so that the “BS approximate” posterior mean of X is given by $E\{X|\mathbf{w}\} \cong \sum_{j=1}^N E\{X|\alpha_j, \beta_j, \theta_j\}/N$.

5. Numerical application

Let now consider the sliding metal wear data given in Meeker and Escobar (1998) and in Ebrahem and Higgins (2006) (see Figure 1, where the data points are connected with straight lines to improve readability). These data refer to an experiment conducted to test the wear resistance of a particular metal alloy, and represent the scar width (in microns, μm) caused by sliding metal wear for an applied weight of 100 grams. The scar width was measured on $m = 4$ specimens at common ages t_k ($t_{1,k} = \dots = t_{4,k} \equiv t_k$, $k = 1, \dots, 8$), up to $T_1 = \dots = T_4 = 500$ hundreds of cycles (see Table 1).

Technical information on the sliding wear phenomenon suggests that the mean wear is a downward concave function, so that the power-law parameter β of the age function $\eta(t) = (t/\alpha)^\beta$ should be less than 1. Since no further information is available on β , the vague Beta prior (20) with parameters $p = q = 0.5$ is chosen.

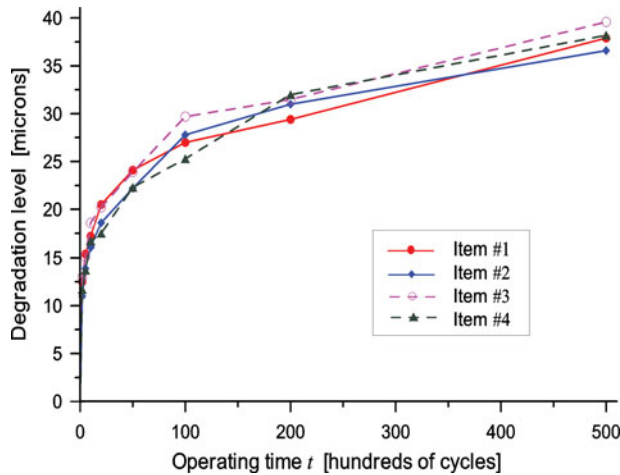


Figure 1. The wear paths in the numerical application.

We also assume that the analyst is able to anticipate a prior mean and a standard deviation on θ , say $E\{\theta\} = 0.1 \mu\text{m}$ and $\sigma\{\theta\} = 0.05 \mu\text{m}$, from which the parameters of the inverse gamma prior (17) are: $a = (E\{\theta\}/\sigma\{\theta\})^2 + 2 = 6$ and $b = E\{\theta\}(a - 1) = 0.5 \mu\text{m}$. Finally, we assume that a prior information on the mean degradation level E_ω at the time $\omega = 300$ hundreds of cycles is available, and that it is formalized in terms of the prior mean $E\{E_\omega\} = 30 \mu\text{m}$ and standard deviation $\sigma\{E_\omega\} = 3 \mu\text{m}$, from which the parameters of the gamma prior (18) are $c = E^2\{E_\omega\}/\sigma^2\{E_\omega\} = 100$ and $d = \sigma^2\{E_\omega\}/E\{E_\omega\} = 0.3 \mu\text{m}$.

An MCMC approach has been adopted for the inferential procedure by exploiting the slice sampling algorithm implemented in the OpenBUGS software. The assumptions on the priors for the parameters E_ω (for $\omega = 300$ hundreds of cycles), β and θ reported in Section 3 and the values of their mean and standard deviation are mapped into the OpenBUGS style, and hence the prior pdf's take the following forms:

- $E_\omega \sim \text{Gamma}(c, d)$, where the shape and the scale parameters are $c = 100$ and $d = 0.3$, respectively;
- $\beta \sim \text{Beta}(p, q)$, where $p = q = 0.5$ of the vague prior for $\beta < 1$;
- $1/\theta \sim \text{Gamma}(a, b)$, where the shape and the rate parameters are $a = 6$ and $b = 1/2$, respectively (θ has an inverse gamma prior distribution).

The likelihood function (16) translates into the following OpenBUGS model:

$$\delta_{i,k} \sim \text{Gamma}(\eta(t_{i,k}) - \eta(t_{i,k-1}), 1/\theta) \quad i = 1, \dots, 4 \text{ and } k = 1, \dots, 8,$$

where $\eta(t_{i,k}) - \eta(t_{i,k-1}) = (E_\omega/\theta)[(t_{i,k}/\tau)^\beta - (t_{i,k-1}/\tau)^\beta]$ and $1/\theta$ are the shape and rate parameters, respectively, and $\tau = 500$ hundreds of cycles.

The OpenBUGS code is launched by an R (R Core Team 2015) computer code by means of the package r2OpenBUGS, a package adapted to OpenBUGS from r2WinBUGS (Sturtz,

Table 1. The wear data [in microns] of specimens [times are in hundreds of cycles].

Specimen i	$t_1 = 2$	$t_2 = 5$	$t_3 = 10$	$t_4 = 20$	$t_5 = 50$	$t_6 = 100$	$t_7 = 200$	$t_8 = 500$
1	12.5	15.4	17.2	20.5	24.1	27.0	29.4	37.9
2	11.0	13.9	16.1	18.6	22.2	27.8	31.0	36.6
3	13.0	15.1	18.6	20.2	23.9	29.7	31.5	39.6
4	11.7	13.7	16.7	17.5	22.3	25.3	32.0	38.2

Ligges, and Gelman 2005) enabling to call OpenBUGS by R and import the samples generated by it into the R environment. The OpenBUGS output is then analyzed by using the R package coda (Plummer et al. 2006), providing diagnostic tools to monitor MCMC convergence. All the computations are made on a notebook based on an Intel® Core™ i7–4960 HQ CPU@2.6GHz.

Convergence is assessed by both running some diagnostic tests and visual inspection of some diagrams. In particular, the Brooks–Gelman–Rubin diagnostic diagram confirms MCMC convergence after $N = 10^6$ iterations with a burn-in period of 10^5 iterations (3 Markov chains have been used for convergence diagnostics), being the “shrink factor” very close to 1 for our simulations. Indeed, this diagram shows the ratio between and within chain variances; it gives evidence of MCMC convergence when the ratio is close to 1 and the between and within variances stabilize (Brooks and Gelman 1998). Furthermore, the autocorrelation diagnostic tool confirms the convergence and points out that no subsampling of the MCMC output is needed (thinning interval equal to 1) since lag-1 autocorrelation is smaller than 0.05.

The execution time of the OpenBUGS routine to collect $N = 10^6$ three-dimensional vector posterior sample of $(E_\omega, \beta, \theta)$ is about 290 s (the elapsed time includes the discarded 10^5 burn-in iterations), which shows that the proposed Bayesian MCMC procedure is fast to apply.

The posterior means of the process parameters are $E\{\theta|\mathbf{w}\} = 0.328 \mu\text{m}$, $E\{\beta|\mathbf{w}\} = 0.207$, $E\{E_\omega|\mathbf{w}\} = 33.2 \mu\text{m}$, and $E\{\alpha|\mathbf{w}\} = 2.48 \cdot 10^{-7}$ hundreds of cycles, and the corresponding 90% HPD intervals are (0.212, 0.440), (0.187, 0.225), (30.8, 35.4), and $(1.90 \cdot 10^{-11}, 4.69 \cdot 10^{-7})$. For a comparative purpose, the ML estimates are equal to $\hat{\theta} = 0.387 \mu\text{m}$, $\hat{\beta} = 0.206$, $\hat{E}_\omega = 34.3 \mu\text{m}$, and $\hat{\alpha} = 1.06 \cdot 10^{-7}$ hundreds of cycles, and the 90% confidence intervals of θ , β , E_ω , and α , based on the log-normal approximation for the distribution of the ML estimators, are respectively (0.256, 0.586), (0.186, 0.228), (29.6, 39.6), and $(4.95 \cdot 10^{-9}, 2.25 \cdot 10^{-6})$.

According to the MCMC procedure described in Section 4, the posterior predictive pdf $f_{X|w_\tau}(x|\mathbf{w})$ of the conditional remaining life of the 4 specimens, given the individual wear level w_τ at time $\tau = T_i = 500$ hundreds of cycles and $w_{\max} = 42 \mu\text{m}$, are computed. Figure 2 compares the posterior predictive pdf's of items #2 and #3 (the items with the smallest and the largest degradation level at $\tau = 500$ hundreds of cycles, respectively) obtained by using both the exact Cdf (2) and the approximate pdf (4). The plots in Figure 2 show that the proposed

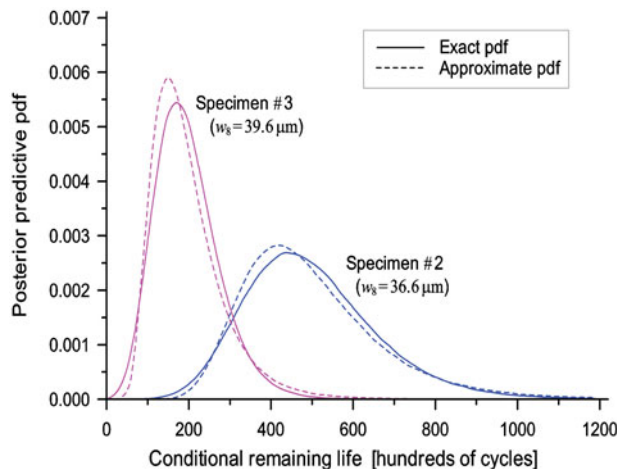


Figure 2. Posterior predictive pdf's of the conditional remaining life of specimens #2 and #3, for $w_{\max} = 42 \mu\text{m}$.

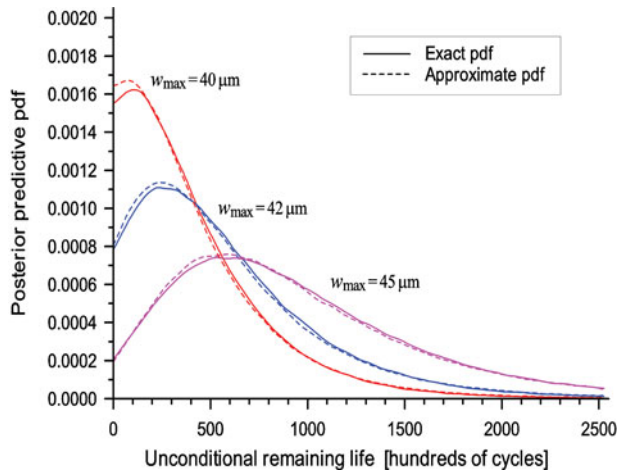


Figure 3. Predictive pdf's of the unconditional remaining life, for $w_{\max} = 40, 42$, and $45 \mu\text{m}$.

approximation works quite well, at least in the range of investigation. For example, the relative error between the approximate and exact estimate of the probability that the conditional remaining life of items #2 and #3 is smaller than 350 hundreds of cycles is equal to 3.2% and -1.9% , respectively. In addition, the plots in Figure 2 highlight the strong influence of the current wear level w_τ on the conditional remaining life.

In Figure 3, the posterior predictive pdf $f_X(x|\mathbf{w})$ of the unconditional remaining life of specimens that are unfailed at time $\tau \equiv t_8 = 500$ hundreds of cycles, obtained by using the approximate pdf (11), is compared to that obtained by using the exact Cdf (8). Selected values of the threshold level, say $w_{\max} = 40, 42$, and 45 microns, are chosen. The non-null value of $f_X(x|\mathbf{w})$ at $x = 0$ is due to the fact that the (unconditional) posterior probability $F_T(\tau|\mathbf{w})$ that a specimen is actually failed at $\tau = 500$ hundreds of cycles is considerably greater than zero, and the smaller $w_{\max}$, the larger $F_T(\tau|\mathbf{w})$. The plots in Figure 3 shows that, in the range of investigation, the proposed approximation (11) for the unconditional remaining life works well. For example, the relative error between the approximate and exact estimate of the probability that the unconditional remaining life is smaller than 1000 hundreds of cycles is equal to -0.5% , 0.04% , and 1.6% , for $w_{\max} = 40, 42$, and 45 microns respectively. It should be highlighted that the execution time required to obtain an approximate posterior sample of the conditional (unconditional) remaining life, starting from the posterior vector of model parameters, is about 0.3 s (26.3 s), that is about 603 times (7 times) smaller than the time required by using the exact Cdf. The very large advantage provided by the proposed approximation of the pdf conditional remaining life is due to the very efficient generation of the BS random elements $z_j|w_\tau$ in (3).

Table 2 gives the exact posterior means of the conditional remaining life $E\{X|w_\tau, \mathbf{w}\}$, given the wear level w_τ at $\tau \equiv t_8 = 500$ hundreds of cycles (see the last column in Table 1), and the unconditional remaining life $E\{X|\mathbf{w}\}$, estimated by averaging the exact posterior sample obtained by using the Cdf (2) and (8), respectively. Those “Exact” conditional and unconditional posterior means are also compared to the “BS exact” means obtained by averaging the posterior samples obtained by using the approximate conditional pdf (4) and unconditional pdf (11), respectively. The “BS approximate” posterior means, obtained by using the approximate expressions (6) and (14), are also provided, in order to check the validity of the suggested approximations. Both the “BS exact” and the “BS approximate” means are quite close to the “Exact” ones, as shown by the corresponding relative errors (which are enclosed in brackets).

Table 2. The exact, the BS exact and the BS approximate posterior mean of the conditional and unconditional remaining life for $\tau = 500$ hundreds of cycles, with $w_{\max} = 42 \mu\text{m}$ (the relative error between the approximate and exact mean are enclosed in brackets).

$w_{\max}$ [μm]		$E\{X w_{\tau}, w\}$ [100 cycles]				$E\{X w\}$ [100 cycles]
		Specimen #1 $w_{\tau} = 37.9$	Specimen #2 $w_{\tau} = 36.6$	Specimen #3 $w_{\tau} = 39.6$	Specimen #4 $w_{\tau} = 38.2$	
40	Exact	170.6	288.0	38.40	145.6	430.6
	BS exact	171.3 (0.41%)	288.7 (0.24%)	39.27 (2.3%)	146.4 (0.55%)	432.0 (0.33%)
	BS approximate	171.2 (0.35%)	288.4 (0.14%)	39.21 (2.1%)	146.2 (0.41%)	427.3 (−0.77%)
42	Exact	357.3	499.3	196.3	326.8	606.7
	BS exact	358.0 (0.20%)	500.0 (0.14%)	197.1 (0.41%)	328.0 (0.37%)	606.0 (−0.12%)
	BS approximate	358.0 (0.20%)	499.9 (0.12%)	196.9 (0.31%)	327.6 (0.24%)	600.0 (−1.1%)
45	Exact	712.2	898.1	499.0	672.2	981.9
	BS exact	713.5 (0.18%)	899.6 (0.17%)	500.1 (0.22%)	673.5 (0.19%)	981.7 (−0.02%)
	BS approximate	713.0 (0.11%)	899.1 (0.11%)	499.9 (0.18%)	673.0 (0.12%)	973.9 (−0.81%)

Indeed, the absolute value of the relative error relative to the conditional mean remaining life ranges from 0.1% and 0.5%, except for few cases where the expected life is very small (specimen #3 and $w_{\max} = 40 \mu\text{m}$, where the relative error is close to 2.3%). In addition, the relative error of the BS exact estimate of the unconditional mean remaining life approximate is, in absolute value, no larger than 0.3%, whereas the relative error of the BS approximation ranges from −0.8% and −1.1% and hence, being the relative error negative, the BS approximation (14) provides conservative estimates of the unconditional mean remaining life. Thus, we can conclude that, at least in the range of investigation, the proposed approximations work well.

The execution time needed to estimate the “BS exact” posterior mean of the conditional remaining life $E\{X|w_{\tau}, w\}$ is very close to that required to estimate the “BS approximate” mean, due to the efficient generation of the BS pseudo-random elements $z_j|w_{\tau}$, and both are about equal to 0.3 s and 600 times smaller than the time required to estimate the “Exact” posterior mean. On the other hand, the time required to estimate the unconditional “BS approximate” mean residual life is about 0.83 s, 220 times smaller than the time required to estimate the unconditional “Exact” mean and about 32 times smaller than the time required to estimate the unconditional “BS exact” mean. Thus, the approximation (14) allows to save much running time, even with respect to the “BS exact” mean.

Table 3 gives the “Exact” and “BS exact” posterior median of the conditional remaining life, given the wear level w_{τ} measured at $\tau = 500$ hundreds of cycles, and the “Exact” and “BS exact” posterior median of the unconditional remaining life. Clearly, due to the asymmetry in the pdf’s of Figures 2 and 3, the posterior median is smaller than the posterior mean. Also

Table 3. The exact and BS exact posterior median of the conditional and unconditional remaining life for $\tau = 500$ hundreds of cycles (the relative error between the approximate and exact median are enclosed in brackets).

$w_{\max}$ [μm]		$X_{0.50} w_{\tau}, w$ [100 cycles]				$X_{0.50} w$ [100 cycles]
		Specimen #1 $w_{\tau} = 37.9$	Specimen #2 $w_{\tau} = 36.6$	Specimen #3 $w_{\tau} = 39.6$	Specimen #4 $w_{\tau} = 38.2$	
40	Exact	162.2	276.0	33.80	137.9	330.6
	BS exact	153.9 (−5.1%)	266.4 (−3.5%)	26.82 (−21%)	130.0 (−5.7%)	322.6 (−2.4%)
42	Exact	343.2	480.2	187.1	313.5	488.9
	BS exact	333.1 (−2.9%)	468.9 (−2.4%)	178.7 (−4.5%)	304.2 (−3.0%)	478.5 (−2.1%)
45	Exact	685.0	864.2	479.9	646.6	833.7
	BS exact	673.2 (−1.7%)	850.6 (−1.6%)	469.3 (−2.2%)	635.0 (−1.8%)	820.8 (−1.5%)

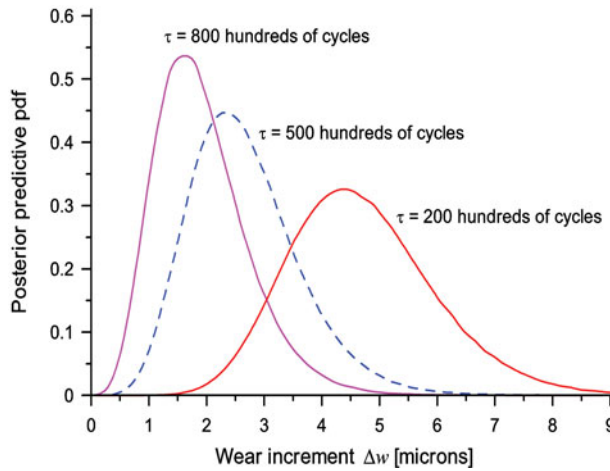


Figure 4. The posterior predictive pdf of the wear increment ΔW over a future interval of width $\Delta t = 200$ hundreds of cycles, for $\tau = 200, 500$, and 800 hundreds of cycles.

the relative errors of the “BS exact” posterior median with respect to the “Exact” one are given in brackets. From the values in Table 3, we have that the proposed approximations work satisfactory, except when the current degradation level is very close to the threshold level (specimen #3 and $w_{\max} = 40 \mu\text{m}$). The execution time required to estimate the conditional or the unconditional “Exact” median, starting from the posterior sample of the model parameters, is about 184 s, and is 560 times (7 times) greater than the time required to estimate the conditional (unconditional) “BS exact” median.

Furthermore, Figure 4 depicts the posterior predictive pdf $f_{\Delta W}(\delta|\mathbf{w})$ in (28) of the wear increment $\Delta W = W(\tau + \Delta t) - W(\tau)$, for $\Delta t = 200$ hundreds of cycles and selected values of the current time τ , say $\tau = 200, 500$, and 800 hundreds of cycles. The corresponding mean values are $1.92 \mu\text{m}$, $2.67 \mu\text{m}$ and $4.71 \mu\text{m}$, while the 90% HPD intervals are $(0.639, 3.12)$, $(1.12, 4.13)$ and $(2.59, 6.70)$. Since the age function $\eta(t)$ is concave, than the larger τ , the smaller (in the usual stochastic order) the wear increment.

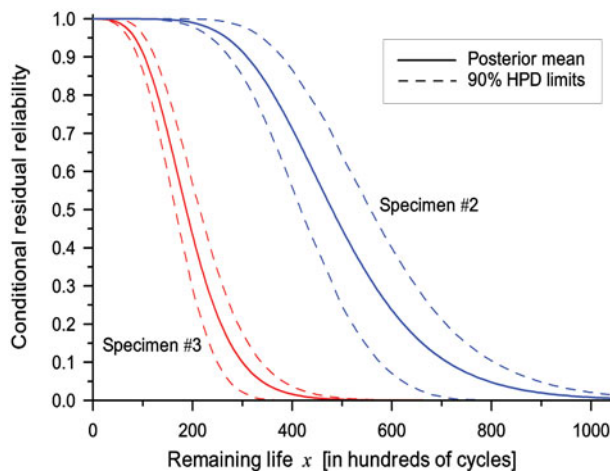


Figure 5. Posterior mean (solid line) and 90% HPD intervals (dashed line) of the conditional residual reliability $R_\tau(x|w_\tau)$ of specimens #2 and #3 at time $\tau = 500$ hundreds of cycles, with $w_{\max} = 42 \mu\text{m}$.

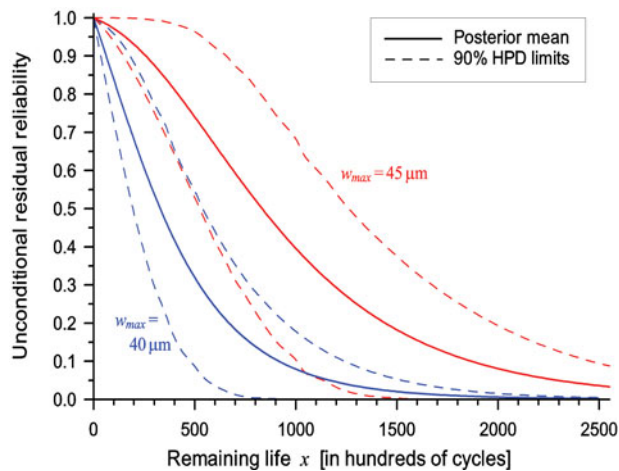


Figure 6. Posterior mean (solid line) and 90% HPD interval (dashed line) of the unconditional residual reliability $R_\tau(x)$ at time $\tau = 500$ hundreds of cycles for $w_{\max} = 40$ and $45 \mu\text{m}$.

Figure 5 gives the posterior mean and the 90% HPD interval of the conditional residual reliability $R_\tau(x|w_\tau)$ of specimens #2 and #3, given the degradation level w_τ at time $\tau = 500$ hundreds of cycles, with $w_{\max} = 42 \mu\text{m}$.

Finally, Figure 6 shows the posterior mean and the 90% HPD interval of the unconditional residual reliability $R_\tau(x)$ at time $\tau = 500$ hundreds of cycles, for $w_{\max} = 40$ and $45 \mu\text{m}$.

6. Conclusions

In this paper, a Bayesian estimation procedure of the parameters of the gamma degradation process has been proposed, when physical/technological prior information on the degradation phenomenon is available. Different types of prior information have been considered and briefly discussed. The posterior distribution of the gamma process parameters, as well as of several quantities of interest such as the residual reliability and the remaining mean life, has been derived. Furthermore, some useful approximations of the mean and median of the remaining life of a technological unit subject to a non-homogeneous gamma degradation process with power-law shape function are derived, both conditioning and not conditioning on the current degradation level.

Point and interval estimation procedures of model parameters and other quantities of interest, based on Markov Chain Monte Carlo techniques implemented in OpenBUGS, have been proposed. Prediction on the degradation increment over a future time interval and on the remaining life of the unit, both conditioning and not conditioning on the current degradation level, has been also illustrated. The proposed Bayesian approach and estimation procedures have been applied to a real case study consisting of the sliding wear of some alloy specimens, thus showing the feasibility of the suggested Bayesian MCMC procedure, the very low computation times, and the goodness of the proposed approximations to the conditional and unconditional remaining life.

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References

- Abdel-Hameed, M. A. 1975. A gamma wear process. *IEEE Transactions on Reliability* 24:152–54. doi:10.1109/TR.1975.5215123.
- Birnbaum, Z. W., and S. C. Saunders. 1969. A new family of life distributions. *Journal of Applied Probability* 6:319–27. doi:10.1017/S0021900200032848.
- Bousquet, N., M. Fouladirad, A. Grall, and C. Paroissin. 2015. Bayesian gamma processes for optimizing condition-based maintenance under uncertainty. *Applied Stochastic Models in Business and Industry* 31:360–79. doi:10.1002/asmb.2076.
- Brooks, S. P., and A. Gelman. 1998. Alternative methods for monitoring convergence of iterative simulations. *Journal of Computational and Graphical Statistics* 7:434–55.
- Çinlar, E. 1980. On a generalization of Gamma process. *Journal of Applied Probability* 17:467–80. doi:10.1017/S002190020004729X.
- Dufresne, F., H. U. Gerber, and E. S. W. Shiu. 1991. Risk theory with the gamma process. *ASTIN Bulletin: The Journal of the International Actuarial Association* 21:177–92. doi:10.2143/AST.21.2.2005362.
- Ebrahim, M. A. H., and J. J. Higgins. 2006. Non-parametric analysis of a proportional wearout model for accelerated degradation data. *Applied Mathematics and Computation* 174:365–73. doi:10.1016/j.amc.2005.05.013.
- Finkelstein, M. S. 2003. A model of aging and a shape of the observed force of mortality. *Lifetime Data Analysis* 9:93–109. doi:10.1023/A:1021886207236. PMID:12602776.
- Greene, W. H. 2003. *Econometric Analysis, fifth ed.* Upper Saddle River, NJ: Prentice Hall.
- Kallen, M. J., and J. M. van Noortwijk. 2005. Optimal maintenance decisions under imperfect inspection. *Reliability Engineering and System Safety* 90:177–85. doi:10.1016/j.res.2004.10.004.
- Lawless, J., and M. Crowder. 2004. Covariates and random effects in a gamma process model with application to degradation and failure. *Lifetime Data Analysis* 10:213–27. doi:10.1023/B:LIDA.0000036389.14073.dd. PMID:15456104.
- Lee, C. E., S. U. Kim, and S. Lee. 2014. Time-dependent reliability analysis using Bayesian MCMC on the reduction of reservoir storage by sedimentation. *Stochastic Environmental Research and Risk Assessment* 28:639–54. doi:10.1007/s00477-013-0779-x.
- Lunn, D., D. Spiegelhalter, A. Thomas, and N. Best. 2009. The BUGS project: Evolution, critique and future directions (with discussion). *Statistics in Medicine* 28:3049–82. doi:10.1002/sim.3680. PMID:19630097.
- Mahmoodian, M., and A. Alani. 2014. Modeling deterioration in concrete pipes as a stochastic gamma process for time-dependent reliability analysis. *Journal of Pipeline Systems Engineering and Practice* 5:1–5. doi:10.1061/(ASCE)PS.1949-1204.0000145.
- Meeker, W. Q., and L. A. Escobar. 1998. *Statistical methods for reliability data.* New York: Wiley.
- Neal, R. M. 2003. Slice sampling. *Annals of Statistics* 31:705–67. doi:10.1214/aos/1056562461.
- Oehlert, G. W. 1992. A note on the delta method. *The American Statistician* 46:27–29.
- Paroissin, C., and A. Salami. 2014. Failure time of non homogeneous gamma process. *Communications in Statistics – Theory and Methods* 43:3148–61. doi:10.1080/03610926.2012.694546.
- Plummer, M., N. Best, K. Cowles, and K. Vines. 2006. CODA: Convergence diagnosis and output analysis for MCMC. *R News* 6:7–11.
- R Core Team. 2015. *R: A language and environment for statistical computing.* Vienna, Austria: R Foundation for Statistical Computing.
- Robert, C. P., and G. Casella. 2010. *Introducing monte carlo methods with R.* New York: Springer.
- Singpurwalla, N. 1985. Survival in dynamic environments. *Statistical Science* 10:86–103. doi:10.1214/ss/1177010132.
- Sturtz, S., U. Ligges, and A. Gelman. 2005. R2WinBUGS: A package for running WinBUGS from R. *Journal of Statistical Software* 12:1–16. doi:10.18637/jss.v012.i03.
- van Noortwijk, J. M. 2009. A survey of the application of gamma processes in maintenance. *Reliability Engineering and System Safety* 94:2–21. doi:10.1016/j.res.2007.03.019.
- Ver Hoef, J. M. 2012. Who invented the Delta method? *The American Statistician* 66:124–27. doi:10.1080/00031305.2012.687494.
- Ye, Z. S., M. Xie, L. C. Tang, and N. Chen. 2014. Semiparametric estimation of gamma processes for deteriorating products. *Technometrics* 56:504–13. doi:10.1080/00401706.2013.869261.